

INUNDATED The Tapkeshwar Mahadev temple in Dehradun, Sept. 16; (inset) the mandir in summer

ANI

UTTARAKHAND

85
DEATH TOLL

₹3,000 cr.
OFFICIAL LOSS ESTIMATE

1,343 mm
AGGREGATE RAINFALL*

22%
EXCESS RAIN

* For the period Jun. 1-Sept. 15, 2025; Source: IMD

THE 2025 FLOODS HAVE EXPOSED MORE THAN CLIMATE FURY, THEY HAVE EXPOSED THE FAULT LINES OF INDIA'S RELIEF SYSTEM. STATE GOVTS COMPETE FOR EMPATHY; THE CENTRE INSISTS ON UNIFORMITY

automated telemetry-based river gauges for real-time monitoring, or dynamic reservoir operation protocols that allow pre-emptive releases based on weather forecasts. Himachal's vulnerability stems from decades of unregulated construction on fragile slopes, deforestation, and poorly safeguarded hydropower projects. The state has yet to implement slope-stabilisation through bioengineering, enforce strict no-construction zones along riverbanks, or instal automatic rainfall–landslide early-warning sensors in high-risk valleys. Amit Prothi, director general, Coalition for Disaster Resilient Infrastructure (CDRI), says, "The challenge today is to make better risk assessments and build resilient infrastructure accordingly."

Uttarakhand would do well to take this advice. The ISRO Landslide Atlas places one of its hill districts, Rudraprayag, as the most landslide-prone in India, followed closely by Tehri Garhwal, while Chamoli, Uttarkashi and Pauri Garhwal districts also figure in the country's top 25 hotspots. The state lies in the Main Central Thrust, a particularly fragile tectonic zone of the Himalayas. The monsoon rains, often intensified by western disturbances, trigger slope failures and cloudbursts in narrow valleys. When these combine with the unplanned construction on riverbeds and floodplains, the damage multiplies. "The Himalayan region now has more structures, more people, more unchecked construction on riverbeds. This can only mean more destruction during freak weather events," says Charu C. Pant, retired professor of geology, Kumaon University.

Experts stress that mitigation cannot be reactive. Long-term resilience must include wetland restoration to absorb excess water, a strengthening of embankments and real-time river monitoring. Coordination between dams and state governments is crucial to manage water release during peak flows. The National Institute of Hydrology, Roorkee, notes that integrated watershed management can reduce flood intensity by 30 per cent if implemented systematically across the northern states. The 2025 deluge will be remembered not just for the lives lost or crops destroyed, but for the uncomfortable truths it revealed: India's rivers, once the cradle of civilisations, are increasingly becoming unpredictable agents of ruin. Whether the lessons are heeded this time will decide if the next great flood comes later, or far sooner. ■

—with Avaneesh Mishra and Kaleem Geelani